Reducing Voting Inequality Through Cost Mechanisms

Evidence from Tempe City Council Elections

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Introduction

Voting is an essential component of political participation and for many Americans, the main way their voice is heard. It ensures that citizens regardless of their social and economic resources wield equal influence in the democratic process. While this may be true in theory, political scientists have known for a long time that political participation is generally unequal. Moreover the inequality is not random but systemic, favoring privileged citizens - those with higher incomes and better education(Lijphart, 2014). This allows special interest groups to disproportionately represent elected officials; A phenomenon that is exacerbated in elections with fewer participants (Steven J. Rosenstone and John Mark Hansen (1993, 238). This proves to be a major elector problem even more so with turnout slowly decreasing in the past few decades(Franklin, 1945). This problem is observed to be the worst in municipal elections with turnout being 50% lower (Berry, 2024) than that of national elections. This suggests that many voters who participate in high-salience national elections fail to participate in lower-salience municipal elections, a phenomenon commonly referred to as voter rolloff. There have been many studies in the past that attempt to understand a voter's decision to vote with Riker and Ordeshook’s (1968) model being widely accepted by the literature. It is as follows, where an individual will vote when $R > 0$.

$$R = PB - C + D$$

P - probability the vote is pivotal - changes the outcome

B - the differential utility derived by a voter from the success of his preferred party

C - Cost of the vote

D - the utility (if any) generated by carrying out his social obligation to vote, i.e., the sociological benefit of voting.

One explanation for voter roll off lies in the cost of voting, which can be broadly divided into direct costs and informational costs. Informational costs can be understood as time it takes to research candidates, and direct costs being the time it takes to perform the action of voting. An individual decides to vote when the perceived utility of voting exceeds the perceived opportunity costs, with direct costs being the biggest barrier (Blais et al., 2019). Municipal elections typically involve higher informational and logistical costs relative to their perceived benefits, as they receive less media coverage and generate lower public awareness. Efforts to lower the cost of voting have expanded across the United States, with Arizona implementing the Active Early Voting List (AEVL), which allows voters to automatically receive mail ballots for each election; such institutions provide a natural setting to examine how changes in voting costs influence political participation. This paper examines how voting costs and socioeconomic factors influence voter roll-off from presidential to municipal elections, with a focus on how cost-reducing mechanisms differentially affect participation across income groups. This contributes to the literature by serving as a starting point for future research on cost-reduction mechanisms that influence voter participation, particularly in understanding how institutional features like AEVL can lower barriers to continued electoral engagement.

Data and Methodologies

The dataset was constructed from 2 cross-sectional datasets - all the voters in the 2024 presidential election from Tempe Arizona, and all those who voted in the 2024 city council elections. By conditioning on individuals who have already demonstrated a baseline willingness to vote, this approach isolates variation in continued participation rather than initial turnout decisions. Each voter was assigned the median income from their respective zip-code, and whether or not they rolled off in the city council elections. Income was binned for every increase in 10k for readability of results, and similarly Age was binned for every increase in 10 years.

The resulting dataset is therefore restricted to presidential voters, with the dependent variable defined as a binary indicator equal to 1 if the individual did not vote in the municipal election (roll-off), and 0 otherwise. This design reframes participation as a conditional decision, allowing the analysis to focus on the marginal costs and incentives associated with lower-salience elections.

The econometric model this paper follows is

$$\boldsymbol{rolloff_i} = \Lambda\left(\beta_0 + \beta_1 AEVL_i + \beta_2 AGE_i + \beta_3 income_z + \beta_4 (AEVL_i \times income_z) + \gamma_z\right)$$

where $\Lambda(\cdot)$ denotes the logistic link function, and γ_z represents ZIP code fixed effect

To estimate the relationship between voting costs and roll-off, I employ a series of logistic regression models of the form:

Model 1: rolloff ~ AEVL

Model 2: rolloff ~ AEVL + AGE

Model 3: rolloff ~ AEVL + AGE + income

Model 4: rolloff ~ AEVL + AGE + AVEL:income

Model 5: rolloff ~ AEVL + AGE + Fixed Effects(ZIPCODE)

where AEVL (Active Early Voting List enrollment) serves as a proxy for the cost of voting, capturing whether an individual automatically receives a mail-in ballot. Age is included to account for life-cycle differences in participation, and ZIP code fixed effects control for unobserved geographic heterogeneity, including neighborhood-level socioeconomic and institutional factors. In some specifications, median income (at the ZIP-code level) is included directly, as well as interaction terms between AEVL and income to test for heterogeneous treatment effects. All models are estimated using logistic regression, with coefficients interpreted in terms of log-odds and reported as odds ratios for substantive interpretation. Standard errors are computed to assess statistical significance. It is important to note that this paper is an observational study looking to find associations between AVEL roll-off and other variables. Below is a summary table of the data.

	Variable	Mean	Mean (%)
0	AEVL	0.823645	82.364532
1	rolloff	0.688818	68.881841
2	voted_city	0.311182	31.118159
3	income_10k	90690.172967	NaN

Results and Interpretation

Variable	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
AEVL	-1.827***	-1.567***	-1.546***	-1.875***	-1.540***
	(0.033)	(0.035)	(0.035)	(0.108)	(0.035)
AGE_std		-1.048***	-1.032***	-1.032***	-1.022***
		(0.010)	(0.010)	(0.010)	(0.010)
income_10k			-0.044***	-0.078***	
			(0.003)	(0.011)	
AEVL:income_10k				0.036***	
				(0.011)	
ZIP Code Fixed Effects	No	No	No	No	Yes
Observations	74095	74095	74095	74095	74095
Pseudo R^2	0.051	0.200	0.203	0.203	0.205

Across the specifications enrollment in AVEL is strongly and consistently associated with reduction in voter roll off. In the baseline model AVEL had the coefficient of −1.827 ($p < 0.001$), indicating a substantial negative relationship with the probability of dropping off in the city election. In odd terms this implies that AEVL voters have approximately

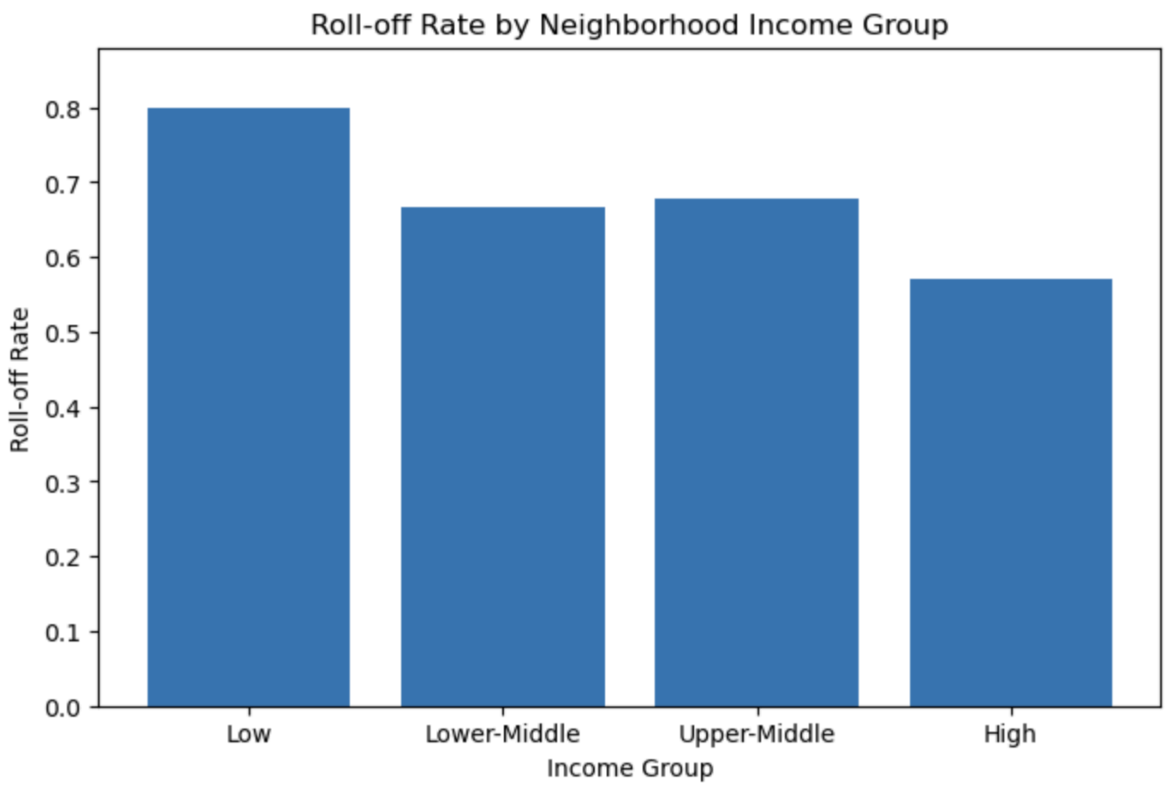
$$e^{-1.827} \approx 0.16$$

Times the odds of rolling off compared to non-AVEL members, holding no other factors constant. This corresponds to an ~84% reduction in the odds of roll-off, suggesting that reducing voting costs is strongly associated with continued participation.

When accounting for age, Pseudo-R^2 goes from 0.02 to 0.208 suggesting that age is a large determinant of an individual's likelihood to roll off. This indicates that older voters are substantially less likely to roll off which is consistent with the literature. This pattern is consistent with prior explanations in the literature; “When individuals get older there is less career pressure and more time to generate interest in politics”(Glenn, & Grimes, 1968).

Importantly, even after controlling for age, the AEVL coefficient remains large (−1.57), suggesting that the association is not fully explained by age differences.

Adding income yields a small but statistically significant effect (−0.044 per $10k, $p < 0.001$), indicating that higher-income individuals are less likely to roll off.



Holding all other variables constant, the relationship between AEVL and voter roll-off exhibits a clear pattern: its association weakens as income increases.

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Income 40k: AEVL effect = -1.731
Income 80k: AEVL effect = -1.587
Income 120k: AEVL effect = -1.443
Income 160k: AEVL effect = -1.299
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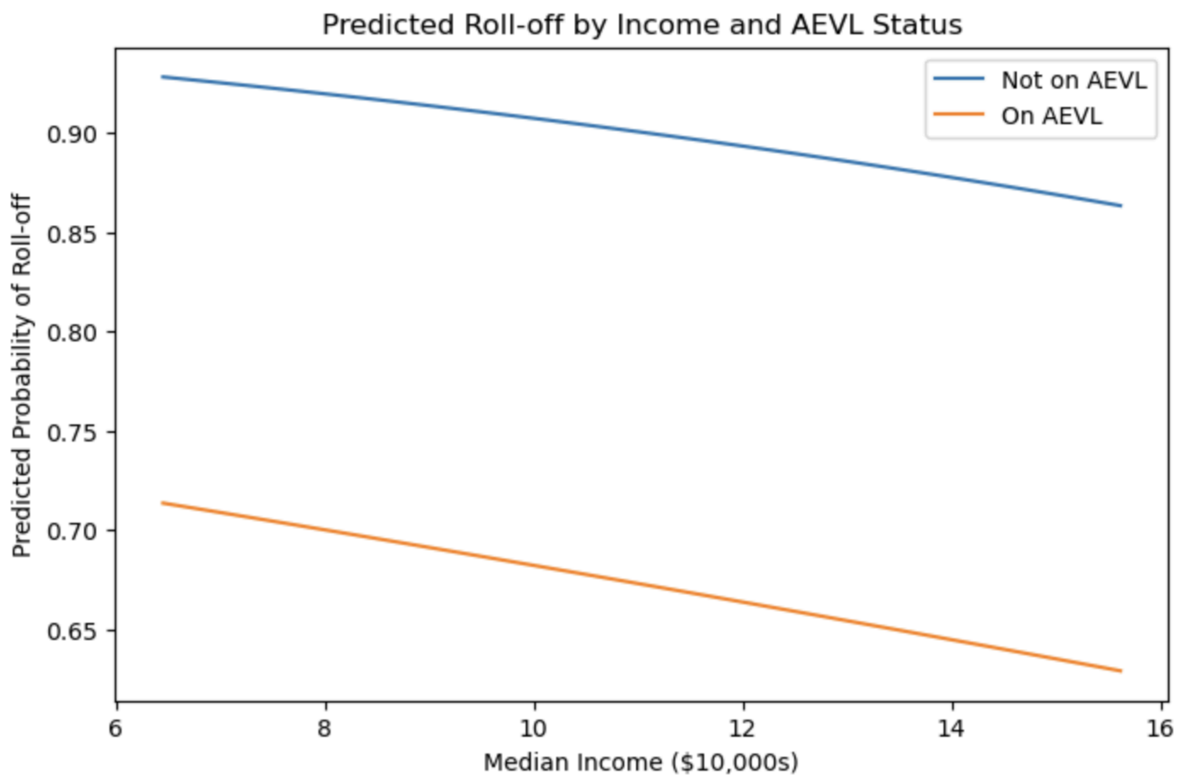
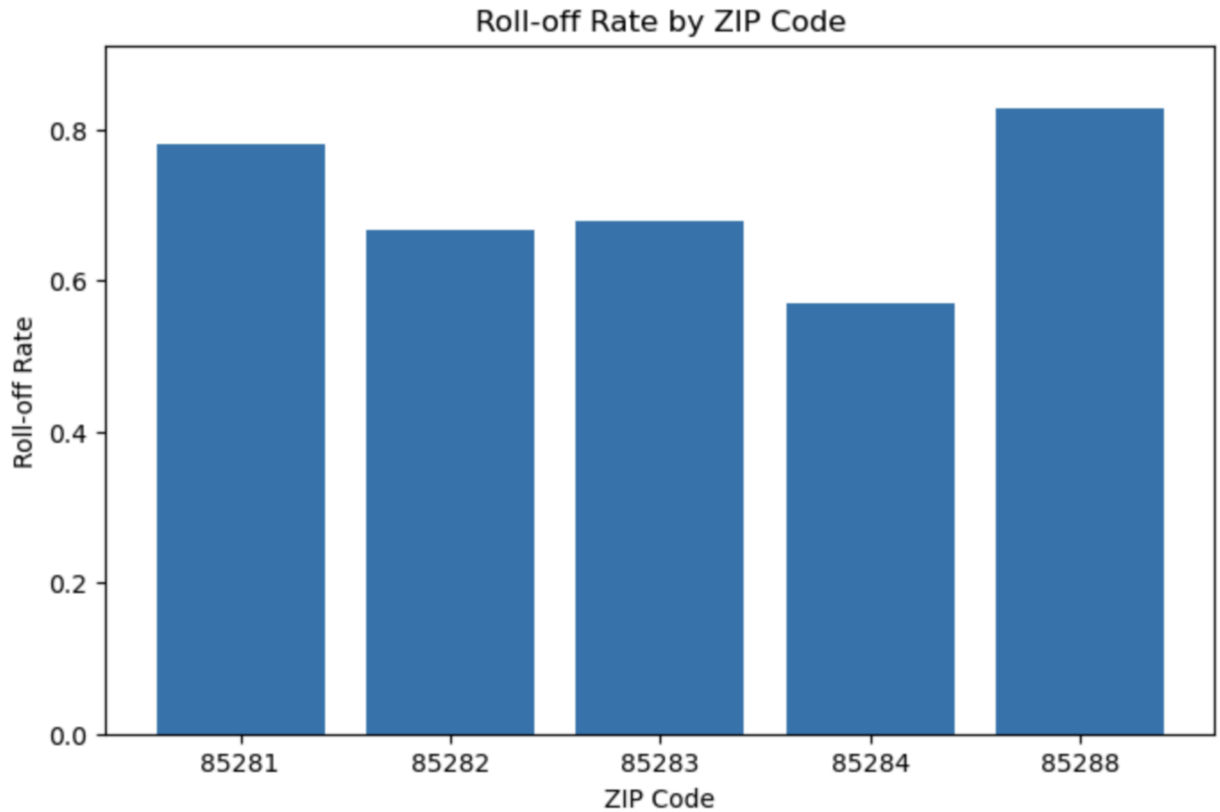


This suggests that cost reducing mechanisms such as AVEL are disproportionately helpful to lower income individuals which helps close the inequality gap in voter turnout. Consistent with the cost based framework, those with higher income have lower barriers to vote, so cost reducing mechanisms have a more marginal effect.

The final model tests with ZIP code fixed effects meaning that voters are compared within zip codes rather than across them. This reduces the geographic and socio-economic effects that could affect voter turnout.



Importantly, the AEVL coefficient remains stable (≈ -1.54) after including ZIP controls, reinforcing that the estimated AEVL effect is not driven by geographic sorting.

Conclusion and Limitations

Across all specifications, AEVL enrollment is associated with a 70–85% reduction in the odds of roll-off, with the effect remaining stable after controlling for demographic and geographic factors. Additionally all models converge successfully, the likelihood ratio tests are highly significant ($p < 0.001$), and Pseudo R^2 increases substantially with added controls suggesting that the model fits well to the data and results are significant. The stability of the AEVL coefficient across models (ranging from −1.83 to −1.54) suggests that the relationship is robust to the inclusion of key covariates. Demographic factors such as age and income are also significant predictors, but their effects are smaller relative to AEVL, indicating that institutional design may be more consequential than individual characteristics in shaping participation decisions. Moreover, the interaction between AEVL and income reveals that cost-reducing mechanisms are most effective among lower-income voters, consistent with a framework in which participation is constrained by unequal access to resources. This suggests that policies aimed at lowering voting costs may also reduce disparities in political participation. While these results provide strong evidence of a robust association between voting costs and roll-off behavior, because enrollment in AVEL is voluntary those who register could already be very interested in politics leading to selection bias. The next step is to obtain data from the years in which AEVL was introduced to run a Difference-in-Differences test to see if the policy shock had any effects. Regardless, this analysis still suggests that voting cost reduction mechanisms like AVEL helps reduce the inequality in voter turnout.

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